

Summarization, Extraction & Generation

Objective

The objective of this practical is to understand three important capabilities of Large Language Models (LLMs):

- Summarization
- Information Extraction
- Text Generation

These tasks help users process information efficiently and improve productivity in education, business, and software development.

1. Introduction

Large Language Models (LLMs) can understand, analyze, and generate human language. Among their many applications, Summarization, Information Extraction, and Text Generation are widely used for reducing lengthy content, extracting important information, and creating new content.

2. Summarization

Definition

Summarization is the process of converting a long piece of text into a shorter version while preserving its key ideas and meaning.

Types of Summarization

A. Extractive Summarization

- Selects important sentences from the original text.
- Keeps the original wording.
- Best for reports and documents.

B. Abstractive Summarization

- Rewrites the content in new words.
- Produces a more natural summary.
- Similar to how humans summarize text.

Example

Original Text

Artificial Intelligence is transforming industries such as healthcare, education, finance, and transportation by improving automation, decision-making, and data analysis.

Summary

Artificial Intelligence improves automation and decision-making across multiple industries.

Prompt Example

Summarize the following article in 100 words. Include: Main idea, Key points, Conclusion.

Applications

- Research paper summaries
- News summaries
- Study notes
- Meeting summaries
- Business reports

3. Information Extraction

Definition

Information Extraction is the process of identifying and extracting specific information from unstructured text.

Common Information Extracted

- Name
- Email
- Phone Number
- Date
- Organization
- Skills
- Location

Example

Input

Rahul Rode joined ABC Technologies on 10 July 2026 as a Data Analyst.

Email: rahul@example.com | Phone: 9876543210

Extracted Information

Field	Value
Name	Rahul Rode
Company	ABC Technologies
Position	Data Analyst
Date	10 July 2026
Email	rahul@example.com
Phone	9876543210

Prompt Example

Extract the following information: Name, Email, Phone Number, Company, Position. Present the result in a table.

Applications

- Resume screening
- Invoice processing
- Customer records
- Medical reports
- Legal documents

4. Text Generation

Definition

Text Generation is the process of creating new content based on a user-provided prompt.

LLMs can generate:

- Emails
- Articles
- Reports
- Stories
- Blog posts
- Programming code
- Product descriptions

Example

Prompt

Write a professional leave application requesting two days of leave.

Sample Output

Subject: Leave Request

Dear Sir/Madam,

I would like to request leave for two days due to personal reasons. I will complete all pending work before my leave.

Thank you for your consideration.

Yours sincerely,

Rahul Rode

Applications

- Email writing
- Content creation
- Documentation
- Chatbots
- Code generation
- Marketing

5. Comparison Table

Feature	Summarization	Extraction	Generation
Purpose	Shorten text	Identify key information	Create new content
Input	Long text	Unstructured text	Prompt
Output	Summary	Structured information	New content
Example	Research paper summary	Resume details	Email writing

6. Advantages

- Saves time
- Improves productivity
- Reduces manual effort
- Quickly processes large documents
- Generates structured outputs
- Supports better decision-making

7. Limitations

- AI may miss important details.
- Generated content should be verified.
- Results depend on prompt quality.
- Long inputs may exceed token limits.

8. Best Practices

- Write clear and specific prompts.
- Provide sufficient context.
- Mention the desired output format.
- Specify word limits when summarizing.
- Review AI-generated content before use.

Viva Questions

Q1. What is Summarization?

Answer: Summarization is the process of shortening a long text while preserving its key ideas.

Q2. What is the difference between Extractive and Abstractive Summarization?

Answer: Extractive summarization selects original sentences from the text, while abstractive summarization rewrites the content using new words.

Q3. What is Information Extraction?

Answer: Information Extraction identifies and extracts specific details such as names, dates, emails, and phone numbers from text.

Q4. What is Text Generation?

Answer: Text Generation is the creation of new content by an AI model based on a user's prompt.

Q5. Name two applications of Information Extraction.

Answer:

- Resume screening
- Invoice processing

Conclusion

Summarization, Information Extraction, and Text Generation are three essential capabilities of Large Language Models. These techniques help users process information efficiently, automate repetitive tasks, and create high-quality content. When combined with effective Prompt Engineering, they improve productivity and support various academic, business, and software development activities.